Week 13 MATLAB App Designer

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Time required: 120 minutes

Reading

Matlab A Practical Introduction to Programming and Problem Solving (Stormy Attaway)

Sections 13.3

Tutorial 1: Create a Calculator with App Designer

MATLAB App Designer is a powerful tool within MATLAB that allows users to create professional apps without needing extensive programming knowledge. It provides a drag-and-drop environment where you can design the user interface (UI) of your app visually and then write code to define its behavior.

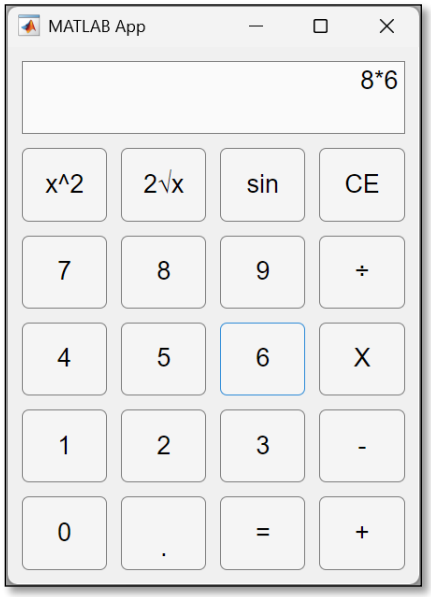
Step 1: Open MATLAB and Start App Designer

1. Open MATLAB.
2. In the **Home** tab, under the **New** button → Click **App.**
3. This should open **App Designer**.

Step 2: Create a New App

1. Click **New** → **Blank App**.
2. Click **Save** → **CalculatorApp**

Step 3: Design the User Interface



1. **Add Components**:
   - Drag and drop a **Grid Layout** from the Component Library to the app.
   - Right Click the Grid Layout → **Add Column**
   - Add **Buttons** for digits (0-9) and operations (+, -, *, /, =).
   - Add a **Text Area** for displaying the input and result.
   - Stretch the Text Area to Fill up all 4 columns.
2. **Arrange Components**:
   - Arrange the buttons in a grid layout to resemble a calculator.
   - Place the text area at the top for displaying the input and result.

Step 4: Set Properties for Components

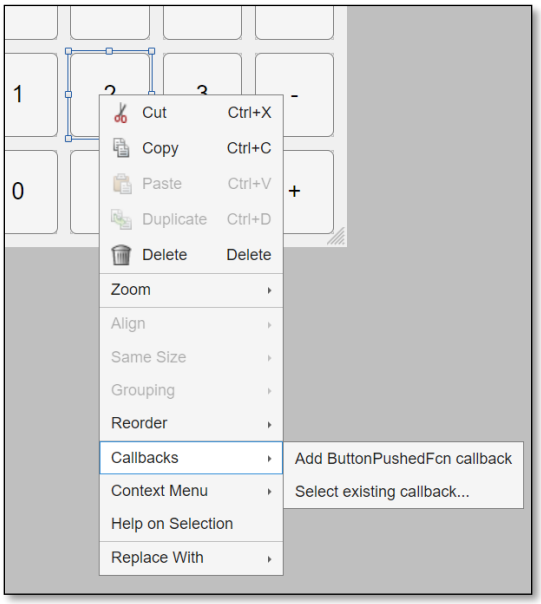
1. **Text Area**:
   - Click the **Text Area** widget.
   - Remove the text in the **Label** box → Press Tab to set the value.

- This will highlight it in the right side Component Browser.
- Right Click the widget → **Rename** → Display
- Press Enter.
- Go to the Interactivity Tool area → set the **Editable** property to false.
- Set the **HorizontalAlignment** to **Right**

2. **Buttons**:
   - Set the **Label** property for each button (e.g., '1', '2', '+', '-', etc.).
   - Rename each button (e.g., Button_1, Button_Plus, etc.).
   - Set the **FontSize** to 18 (Or another size)
   - **Divide** button → hold down the Alt key and type 0247 on the numeric keypad.
     1. You can also copy and paste this symbol ÷

Step 5: Write Callbacks for Buttons

A callback in MATLAB is a function that is executed in response to a specific event or user interaction, such as clicking a button, changing a slider value, or closing a window. Callbacks are essential for creating interactive applications, as they define the behavior of UI components when users interact with them.



1. **Digit Buttons**:

- Right Click on the Button → **Callbacks** → **Add**
- Add the following highlighted code changing the number for each numbered button.

```
% Button pushed function: Button_1
function Button_1Pushed(app, event)
    app.Display.Value = [app.Display.Value, '1'];
end
```

- Click the **Design View** button to return.

3. **Operation Buttons**:
   - Write callback functions for operation buttons [+, -, X, ÷] to append the operation to the display.

```
% Button pushed function: Button_divide
function Button_dividePushed(app, event)
    app.Display.Value = strcat(app.Display.Value, '/');
end
```

4. **Equal Button**:
   - Write a callback function for the equal button to evaluate the expression in the display.

```
% Button pushed function: Button_equals
function Button_equalsPushed(app, event)
        % Ensure the display value is a string scalar
        expression = string(app.Display.Value);

        % Evaluate the expression
        result = eval(expression);

        % Display the result
        app.Display.Value = num2str(result);
end
```

Step 6: Test the App

1. Click on the **Run** button to test your app.
2. Enter numbers and operations to see if the calculator works correctly.

Step 7: Share the App

Share your app by packaging it as a standalone application.

1. Click the **Designer** Tab.
2. Click **Share** → **Standalone Desktop App**
3. Click **Package**. A folder will open when it is done.
4. Go to the **for_redistribution_files_only** folder.
5. Double Click **CalculatorApp**. The program should open.
6. You can delete the other folders and keep the **for_redistribution_files_only** folder if you wish.

Assignment Submission

1. Submit properly named and commented script files.
2. Attach a screenshot of the program showing successful execution.
3. Attach all to the assignment in Blackboard.